

SIMATS ENGINEERING

Department of Computer Science & Engineering

Course Code:	CSA1209 & CSA1215	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL 2-EXPERIMENTS
Weightage:	40%	Total Marks:	25
CO4 Statement:	Design processor organization by constructing pipelined datapath structures, identifying hazard types (data, control, structural), and comparing superscalar techniques such as out-of-order execution, speculative execution, and simultaneous multithreading (SMT). (BL6)		
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 MEM.HIERARCHY

01 Cache02 SRAM/DRAM03 VM/Paging04 NAND/DRAM05 MRAM/RRAM

01

L1, L2, and L3 cache

Q1

Latency, bandwidth, and throughput across the on-chip cache hierarchy, and how each level shapes overall processor performance.

LEVEL	TYPICAL LATENCY	TYPICAL SIZE	BANDWIDTH	SCOPE
L1	4-5 cycles (~1 ns)	32-64 KB, split I/D	Highest — direct core-adjacent ports	Private, per core
L2	10-20 cycles (~3-5 ns)	256 KB-2 MB	High, lower than L1	Private, per core
L3	30-70 cycles (~10-40 ns)	8-64 MB	Lower per-access, higher aggregate	Shared across cores

RELATIVE ACCESS LATENCY

L1 ~1 ns

L2 ~4 ns

L3 ~25 ns

DRAM (for reference) ~90 ns

IMPACT ON PERFORMANCE

Because each level is smaller and faster than the one below it, the hierarchy lets a processor exploit temporal and spatial locality: most accesses hit in L1, so the effective average memory access time stays close to L1 latency even though DRAM is two orders of magnitude slower. Effective access time is a weighted sum — hit rate at each level times its latency, plus the miss penalty of falling through to the next level. A small drop in L1 hit rate is expensive because every miss pays the full cost of an L2 or L3 lookup; a well-tuned cache hierarchy is what keeps a multi-GHz core from spending most of its cycles waiting on memory.

02

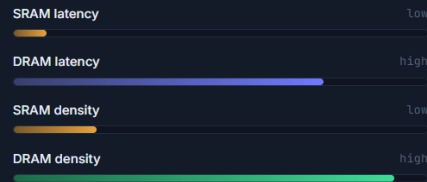
SRAM vs. DRAM

Q2

The two workhorse memory technologies, compared on latency, bandwidth, and throughput as used in cache and main-memory design.

PROPERTY	SRAM	DRAM
Cell structure	6-transistor flip-flop	1 transistor + 1 capacitor
Latency	~1-2 ns	~50-100 ns
Refresh	None — stable while powered	Required every ~64 ms
Bandwidth	Very high, wide on-die ports	High aggregate via wide buses / channels
Density / cost per bit	Low density, expensive	High density, cheap
Typical role	Registers, L1-L3 cache	Main memory (DIMMs)

LATENCY VS. DENSITY TRADE-OFF



WHY BOTH EXIST

SRAM's speed comes directly from its cell design — no capacitor to sense or refresh means near-instant, stable reads, but six transistors per bit makes it too

03

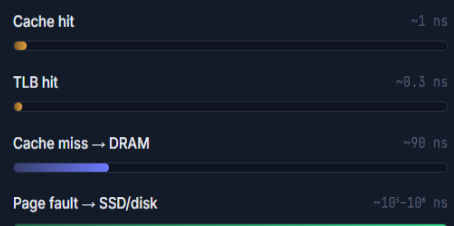
Virtual memory & paging vs. cache

Q3

Two address-translation and staging mechanisms — one hardware-managed, one OS-managed — compared on access latency and system throughput.

PROPERTY	CACHE MEMORY	VIRTUAL MEMORY / PAGING
Managed by	Hardware (cache controller)	OS + MMU / TLB hardware assist
Unit moved	Cache line (~64 B)	Page (typically 4 KB)
Hit latency	~1-40 ns	TLB hit: ~1 cycle
Miss path	Next cache level → DRAM	Page walk, or page fault → disk/SSD
Miss penalty	Tens to ~100 ns	Microseconds to milliseconds
Visibility to software	Fully transparent	Transparent addressing, OS-visible faults

MISS PENALTY, LOG SCALE



04

Q4

NAND Flash vs. DRAM

Latency, bandwidth, and where each technology fits in a hierarchical memory system spanning memory and storage.

PROPERTY	DRAM	NAND FLASH
Addressing	Byte-addressable	Page / block-addressable
Read latency	~50-100 ns	~25-100 µs
Write / program latency	~50-100 ns	~200-500 µs, block erase in ms
Volatility	Volatile — needs power	Non-volatile
Endurance	Effectively unlimited	Limited (~10 ³ -10 ⁵ program/erase cycles)
Density / cost per bit	Lower density, costlier	Very high density, cheap
Hierarchy role	Main memory tier	Persistent storage tier

